

Coupling interface - design brief

AI BRAKING / AB-COUP / C0 / 2026-09-07

Two flanged hubs and a spacer envelope explore a bought flexible element. Tooth forms and elastomer geometry are not designed, so this is not a working gear or flexible coupling.

TARGETS (not validated)

Illustrative shaft bores: 50 mm. Unreleased shaft fit

Hub flange diameter: 160 mm. Concept geometry

Six-hole pattern: PCD 120 mm, Ø11. M10 candidate

Torque / misalignment / speed: Unassigned. Supplier and shaft-system study required

DESIGN DECISIONS

Buy the flexible element or full coupling for the first prototypes.

Keep hub removal and brake-disc access visible in the packaging study.

Treat torsional resonance and misalignment separately from static torque.

CANDIDATE MATERIALS

Hubs: 42CrMo4 candidate, condition to be selected for section and fatigue.

Flexible element: vendor-controlled compound or gear assembly.

Assembly and engineering gates

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ASSEMBLY INTENT

- 1. Approve shaft and torsional system requirements.
- 2. Replace spacer with qualified supplier geometry.
- 3. Finalize hub attachment, alignment and guarding before prototype assembly.

OPEN DESIGN RISKS

- 1. No transmitting connection between the placeholder flexible element and hubs is released.
- 2. No keyway, shaft fit, balancing or retention design.

VALIDATION REQUIRED

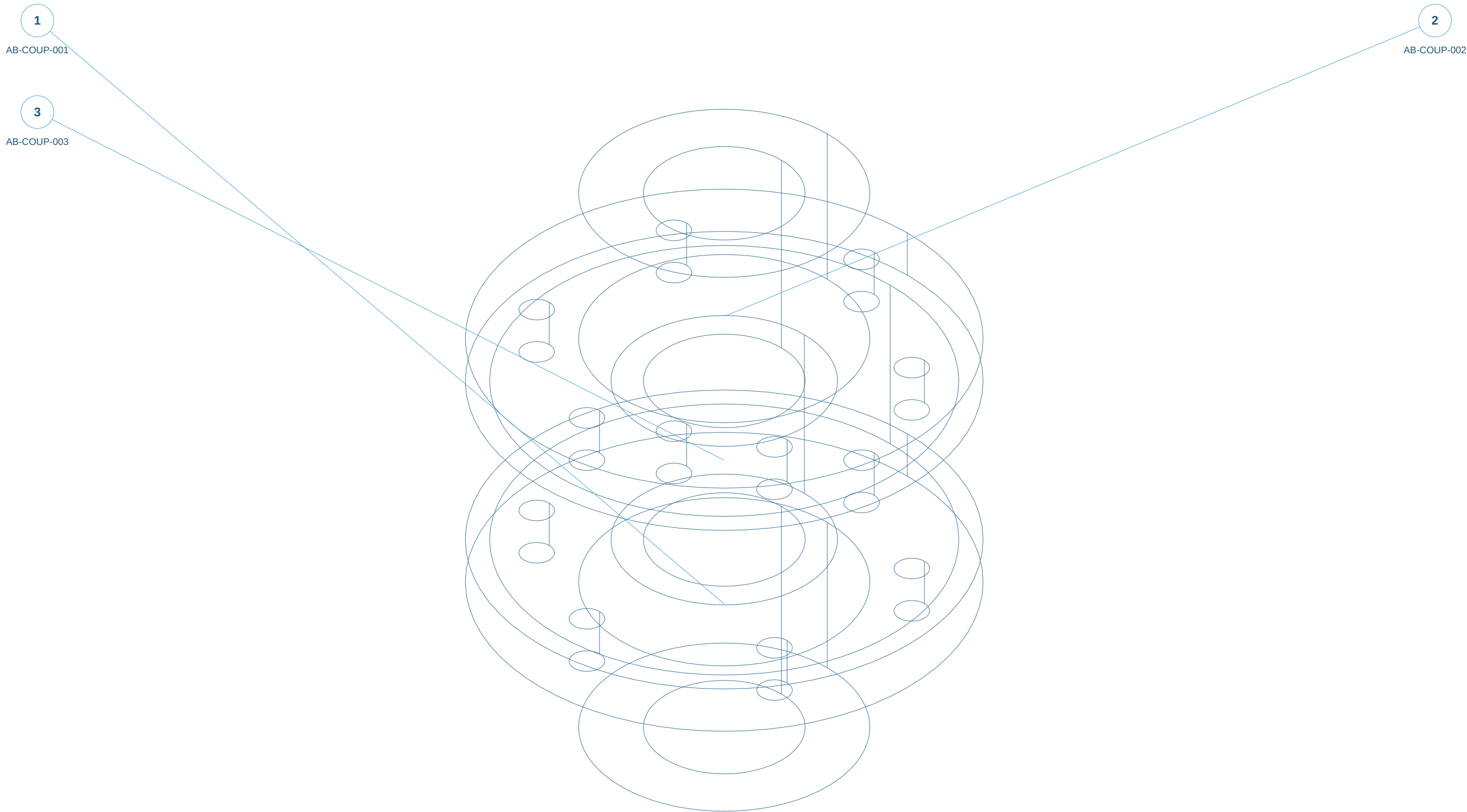
- 1. Shaft torsion, key/locking connection, fatigue and critical-speed assessment.
- 2. Supplier torque/misalignment endurance and balance confirmation.

ELECTRICAL / SENSOR REQUIREMENTS ONLY

- 1. Reserve guard clearance and optional vibration/speed monitoring; no sensors in rotating masses.

Assembly reference / numbered components

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Static isometric reference with item marks. Exploded geometry and part selection are available in the web workspace.

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

Parts and interfaces

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AB-COUP-001 | Hub -1 | make-concept

42CrMo4 candidate; heat treatment / fit unapproved; Saw/laser blank; CNC critical faces and bores

AB-COUP-002 | Hub 1 | make-concept

42CrMo4 candidate; heat treatment / fit unapproved; Saw/laser blank; CNC critical faces and bores

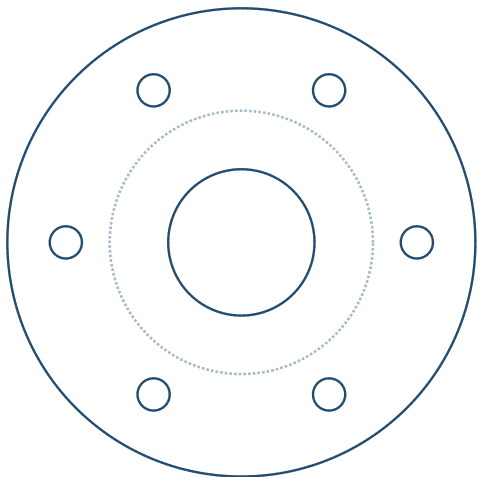
AB-COUP-003 | Flexible-element reserved envelope | interface-envelope

Supplier-controlled; illustrative reserved envelope; Buy complete coupling or engineered flexible element

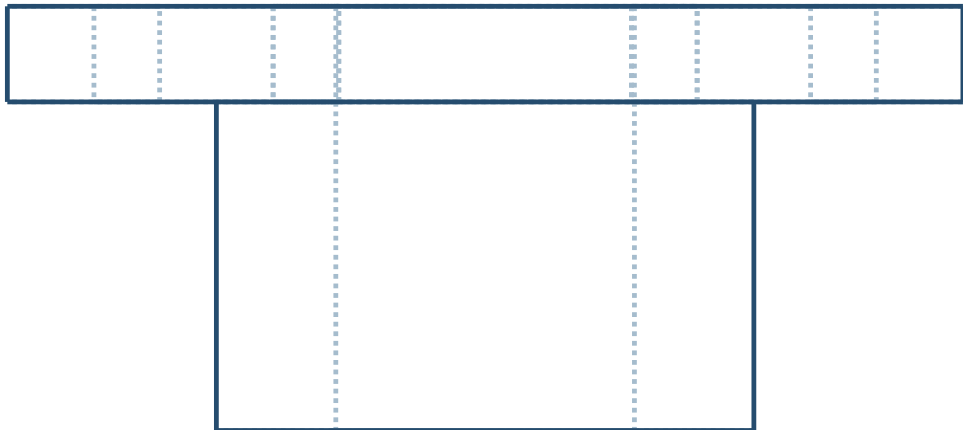
AB-COUP-001 / Hub -1

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TOP X-Y



FRONT X-Z



42CrMo4 candidate; heat treatment / fit unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 160.00 / 160.00 / 71.00 mm

Flange OD 160; bore 50; six diameter 11 on PCD 120.

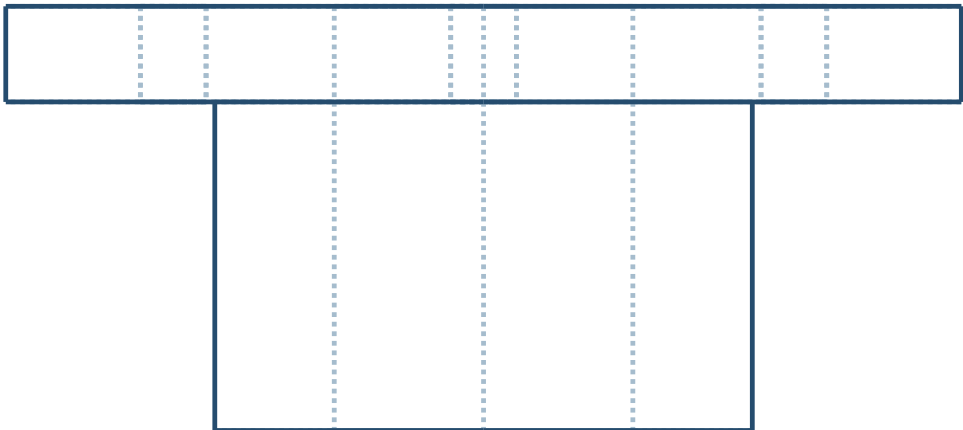
No shaft key/locking or torque-transmitting flexible interface.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

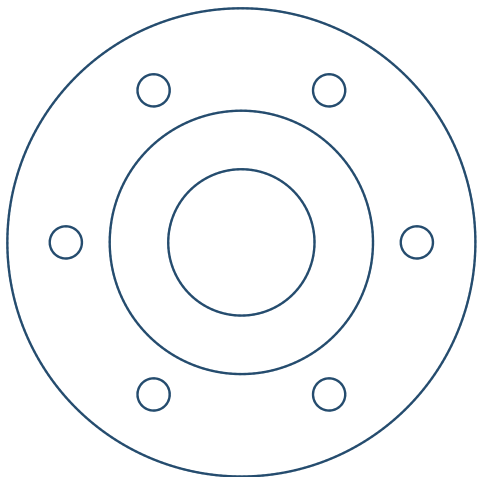
SIDE Y-Z



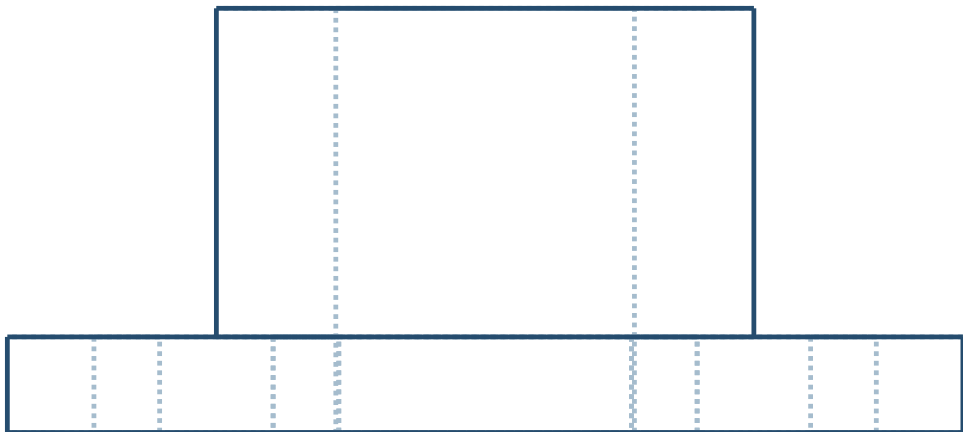
AB-COUP-002 / Hub 1

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TOP X-Y



FRONT X-Z



42CrMo4 candidate; heat treatment / fit unapproved

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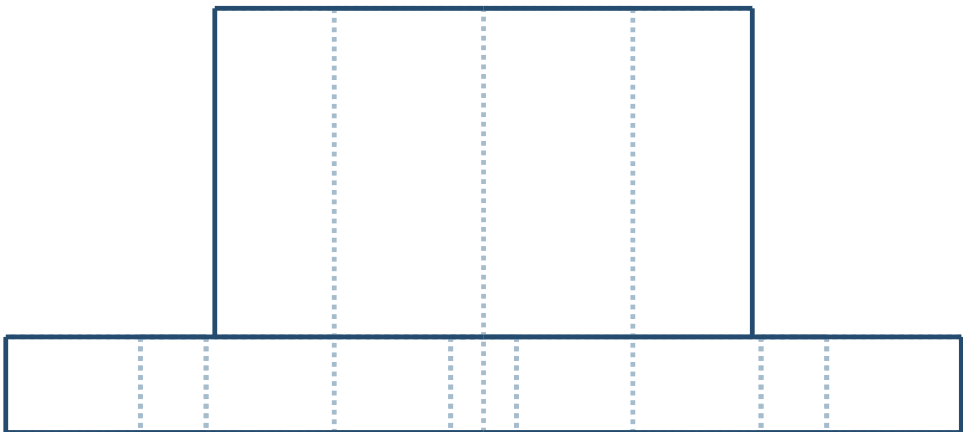
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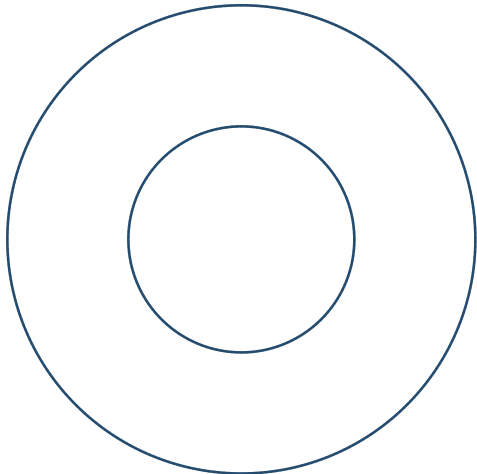
SIDE Y-Z



AB-COUP-003 / Flexible-element reserved envelope

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Buy complete coupling or engineered flexible element

Overall X/Y/Z bounds: 145.00 / 145.00 / 60.00 mm

Reserved space, not a functioning elastic or gear mechanism.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z

